

SCATTERPLOTS, CORRELATION & LSRL

new 5.1–5.3, 5.5 • old 2.4–2.6, 2.8

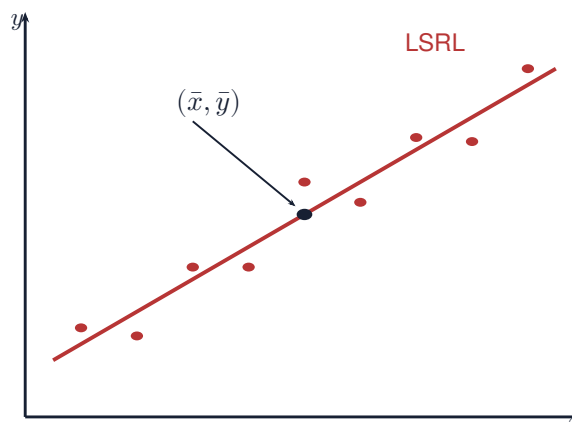
ZONE A · THE LEAST-SQUARES REGRESSION LINE

$$\hat{y} = a + bx$$

$$b = r \frac{s_y}{s_x} \quad a = \bar{y} - b\bar{x}$$

$$\bar{y} = a + b\bar{x}$$

The LSRL always passes through $(\bar{x}, \bar{y})$. It minimizes the sum of squared vertical residuals.



$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

DIRECTION · FORM · STRENGTH · UNUSUAL POINTS

Describe the association using all four, with the variables and their context.

ZONE B · WHAT THE SYMBOLS MEAN + HOW r BEHAVES

x : explanatory variable

y : observed response

$\hat{y}$: **predicted** response

a : intercept; b : slope

s_x, s_y : sample SDs; n : number of pairs

$-1 \leq r \leq 1$; **r has no units.**

Swapping x and y leaves r unchanged.

Changing measurement units leaves r unchanged.

r measures **linear** direction and strength only.

Outliers can change r substantially.

ZONE C · SAY IT IN WORDS

SLOPE

"For each additional _____ [unit of x], the predicted _____ [y] [increases/decreases] by _____ [b] _____ [units of y]."

INTERCEPT

"When _____ [x] is 0, the predicted _____ [y] is _____ [a] _____ [units]."

This may not make sense in context.

ZONE D · WATCH OUT

Correlation does not establish causation.

Inspect the scatterplot and residual plot (P15) for curvature, even when $|r|$ is large.

Slope has units of y per unit of x ; r has no units.